SOCIALMIND DRIVE

Independent components. Shared research theme: safer autonomous mobility under uncertainty.

Problem → gap/objectives → component architecture → evidence/NFRs → demo roadmap → value



SLIIT
FACULTY OF COMPUTING

AGENTIC AI FOR ROBUST AND SOCIALLY-AWARE AUTONOMOUS DRIVING



Agentic AI for Robust and Socially Aware Autonomous Driving

Component	How the Comments Were Addressed	PP1 Comments
C2	The system was redesigned as a hybrid approach, using deterministic traffic rules for clear right-of-way cases and MAPPO-based multi-agent reinforcement learning for ambiguous or unresolved negotiations.	Avoid relying solely on rule-based decision making
C3	The cultural adaptation component was removed from the system. Consequently, the proposed framework no longer incorporates region-specific recalibration or Sri Lanka-specific threshold adjustments.	Remove Cultural Adaptation
C4	The approach was revised so that raw sign images remain on each vehicle. Instead, the detected sign is converted into an embedding, and only this embedding is shared between vehicles for collective learning.	Sharing raw traffic-sign images between vehicles is not aligned with Federated Learning principles

1. Problem, stakeholders and measurable pain point

Autonomous vehicles fail most often when the road scene is uncertain, incomplete, or rare.

Stakeholders: AV developers, road-safety researchers, transport authorities, fleet operators, and vulnerable road users.

Occluded pedestrian intent

Partial visual evidence can make crossing intention unreliable.

Unclear right-of-way

Traffic rules alone may not provide a clear crossing order.

Lack of Hidden Pedestrian Mental-State Inference

Most systems are reactive, detecting crossing behavior only after the pedestrian commits, without considering the underlying mental state

Limited collective learning

Vehicles learn mainly from local data and cannot easily benefit from rare experiences of other vehicles.

Research objective

Build independent AI agents that can predict, reason, explain, negotiate, or share knowledge under uncertainty — without hiding safety behind a black-box controller.

2. Research gap and objectives

Important scope statement

The components share an AV safety theme and presentation/demo layer, but they are not runtime inputs to each other.

C1

Occluded pedestrian intent

Binary intent prediction is risky when visibility is unreliable.

Defer unreliable decisions

C2

Right-of-way negotiation

Unsignalized intersections can create unresolved precedence.

Resolve conflict order safely

C3

Theory-of-Mind pedestrian agent

Classifiers often lack human-readable driving rationale.

Proactive action

C4

Agentic Knowledge Sharing

Vehicles learn from limited local data, making rare or new signs difficult to recognize.

Selectively share verified sign knowledge for collective learning.

Four technical pipelines, one research story.

Each component has its own data source, model, decision layer and demo proof point.

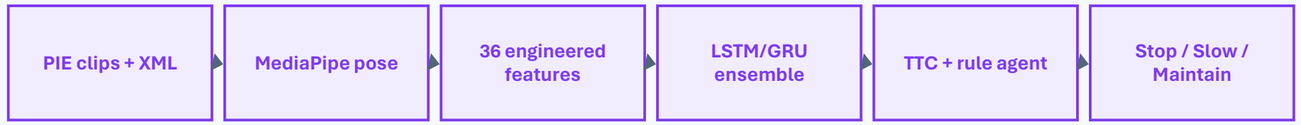
C1 Occlusion-aware pedestrian intent



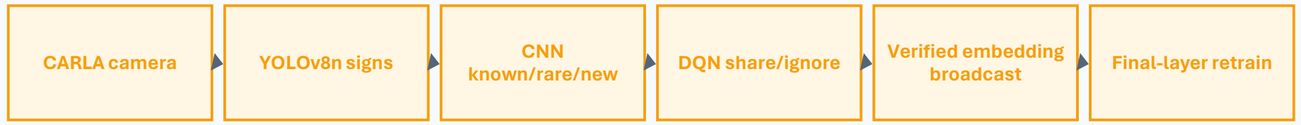
C2 AV right-of-way negotiation



C3 Theory-of-Mind pedestrian agent



C4 Agentic Knowledge Sharing



4. Evaluation evidence, metrics and current limitations

Intent Prediction Performance

Metric	Result
Accuracy	**87.59%**
Precision	61.63%
Recall	75.50%
F1-Score	67.87%
ROC-AUC	88.28%
Average Precision	69.28%

Selective Decision Performance

Metric	Result
Committed Accuracy	**88.94%**
Coverage	**87.15%**
Deferred / Observe More	12.85%
Base Intent Accuracy	87.59%

Automatic Perception Performance

Component	Metric	Result
YOLO Detector	mAP@0.5	30.6%
YOLO Detector	Precision	43.3%
YOLO Detector	Recall	38.1%
Occlusion Estimator	Accuracy	58.0%
Occlusion Estimator	Macro F1	57.8%

Current Limitations

- Pedestrian detection recall remains moderate in difficult and heavily occluded scenes.
- Tracking stability decreases during severe or prolonged pedestrian occlusion.
- Automatic occlusion estimation still has room for improvement.
- A monocular RGB system cannot detect a completely hidden pedestrian who has never previously been observed.

Further evaluation across additional datasets and simulated urban scenarios is required.

4. Evaluation evidence, metrics and current limitations

Multi-Agent Negotiation for Right-of-Way in Complex Intersections

GRU Intention Prediction

Horizon	GRU Accuracy	Logistic Accuracy	GRU Macro F1
1.0 s	**89.66%**	79.31%	**88.93%**
0.5 s	**93.10%**	87.93%	**92.38%**

MAPPO Model Selection

Candidate	PPO Epochs	Mean Validation TTT	Safety Gate	Result
E5	**5**	**15,817.91 s**	Passed	**Selected**
E10	10	16,579.07 s	Passed	Not selected
E15	15	16,942.56 s	Passed	Not selected

Final Held-Out Evaluation

Metric	Selected MAPPO E5
Held-out scenarios	36
Replications	3
Scenario executions	108
Collisions	**0**
Blocked-zone violations	**0**
Native SUMO safety interventions	**0**
Total Team Travel Time	**46,077.48 s**
Mean scenario TTT	**426.64 s**

GRU Intention Prediction — Test Results

Prediction horizon	Model	Test Accuracy	Balanced Accuracy	Macro F1
1.0 s Primary	**GRU-32**	**89.66%**	**89.10%**	**88.93%**
1.0 s	Weighted Logistic	79.31%	76.60%	77.46%
0.5 s Secondary	**GRU-32**	**93.10%**	**91.67%**	**92.38%**
0.5 s	Weighted Logistic	87.93%	86.22%	85.91%

Multi-Agent Negotiation for Right-of-Way in Complex Intersections

Limitations

- Simulation-based validation only
- Limited intersection and traffic-rule scope
- Bounded MAPPO model-selection experiment

4. Evaluation evidence, metrics and current limitations

Proactive Mental State Prediction Modeling using Theory of Mind

LSTM - Class-wise Metrics

Class	Precision	Recall	F1
Waiting	39%	45%	42%
Hesitant	74%	72%	73%
Committed	85%	78%	81%
Distracted	22%	39%	28%
Aggressive	75%	81%	78%
Jaywalk	58%	53%	55%

GRU - Class-wise Metrics

Class	Precision	Recall	F1
Waiting	39%	58%	47%
Hesitant	77%	72%	75%
Committed	87%	75%	81%
Distracted	28%	54%	37%
Aggressive	75%	54%	80%
Jaywalk	70%	47%	56%

Ensemble - Class-wise Metrics

Class	Precision	Recall	F1
Waiting	40%	49%	44%
Hesitant	79%	73%	76%
Committed	87%	78%	82%
Distracted	25%	51%	34%
Aggressive	76%	84%	80%
Jaywalk	63%	51%	56%

Overall Test Performance

Model	Accuracy	Macro F1
LSTM	69.80%	59.50%
GRU	70.80%	62.60%
Ensemble	**71.50%**	**62.10%**

Limitations

- Class imbalance reduces reliable prediction of rare mental states such as Distracted and Waiting
- Limited real-world validation - evaluation mainly relies on the PIE dataset and CARLA simulation
- Limited computational resources restricted extensive hyperparameter tuning and model experimentation

4. Evaluation evidence, metrics and current limitations

Federated Collective Learning Between Autonomous Vehicles using Agentic Knowledge Sharing

- .Vehicle A trained mainly on Prohibitory signs, Vehicle B on Mandatory signs, Vehicle C on Danger signs
- Each vehicle tested across all 3 categories, using shared test set
- Evaluated across multiple test stages (Test 0 + Test 3): Test 0 = before knowledge sharing, later tests = after each sharing + retraining round

Vehicle A (Class-wise Accuracy)

Class	Test 0	Test 1	Test 2	Test 3
0	87.50%	87.50%	75.00%	87.50%
1	28.57%	57.14%	71.43%	57.14%
4	37.50%	87.50%	87.50%	87.50%
7	80.00%	70.00%	70.00%	90.00%
13	0.00%	100%	100%	87.50%
14	22.22%	100%	100%	77.78%
18	0.00%	80.00%	80.00%	100%
23	0.00%	77.78%	88.89%	66.67%
27	100%	0.00%	0.00%	85.71%
33	0.00%	16.67%	33.33%	66.67%
35	60.00%	60.00%	60.00%	60.00%
40	0.00%	50.00%	75.00%	100%

Vehicle B (Class-wise Accuracy

Class	Test 0	Test 1	Test 2	Test 3
0	25.00%	75.00%	75.00%	75.00%
1	14.29%	42.86%	42.86%	71.43%
4	0.00%	50.00%	50.00%	50.00%
7	20.00%	60.00%	70.00%	70.00%
13	0.00%	100%	0.00%	100%
14	22.22%	88.89%	88.89%	88.89%
18	0.00%	70.00%	100%	100%
23	0.00%	77.78%	77.78%	66.67%
27	28.57%	85.71%	28.57%	100%
33	0.00%	0.00%	71.43%	71.43%
35	80.00%	80.00%	80.00%	80.00%
40	75.00%	50.00%	75.00%	75.00%

Vehicle C (Class-wise Accuracy)

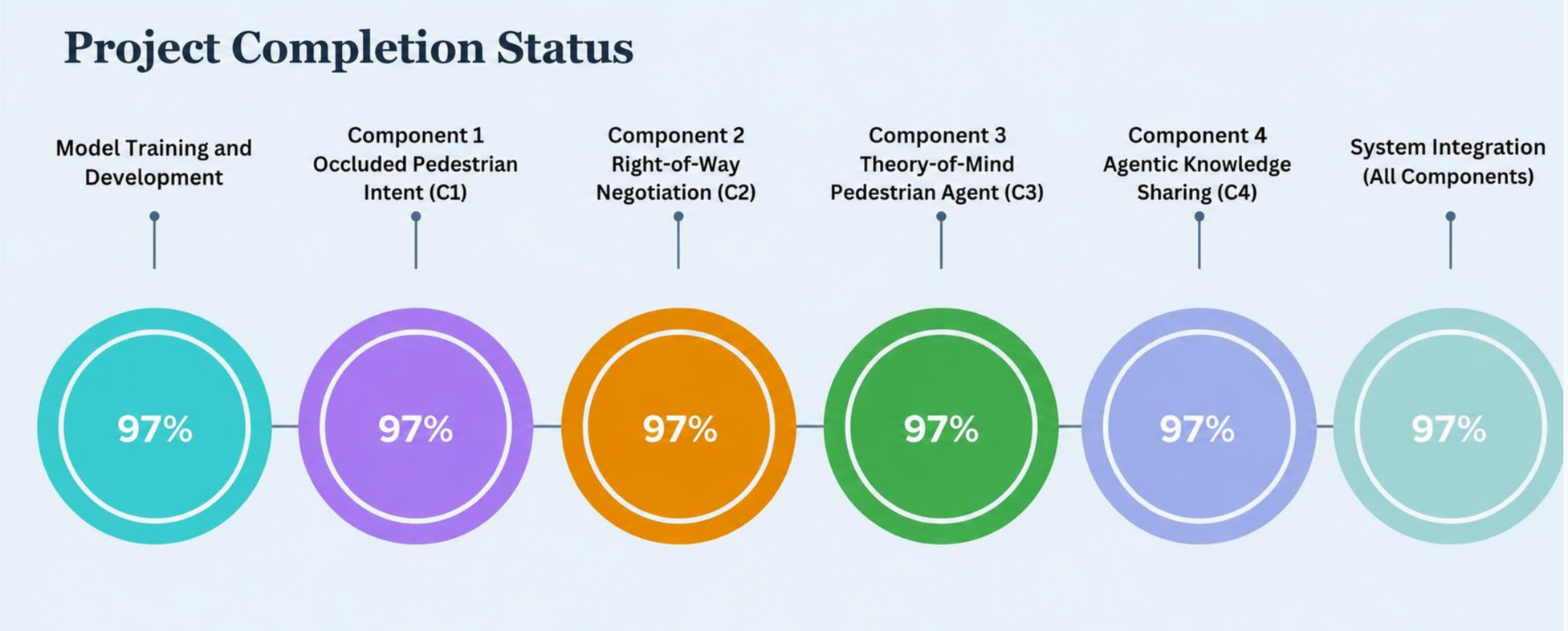
Class	Test 0	Test 1	Test 2
0	0.00%	50.00%	75.00%
1	100%	42.86%	85.71%
4	0.00%	75.00%	50.00%
7	20.00%	70.00%	20.00%
13	25.00%	100%	100%
14	55.56%	88.89%	77.78%
18	70.00%	100%	90.00%
23	55.56%	88.89%	88.89%
27	85.71%	57.14%	100%
33	0.00%	33.33%	66.67%
35	40.00%	100%	60.00%
40	0.00%	75.00%	75.00%

Federated Collective Learning Between Autonomous Vehicles using Agentic Knowledge Sharing

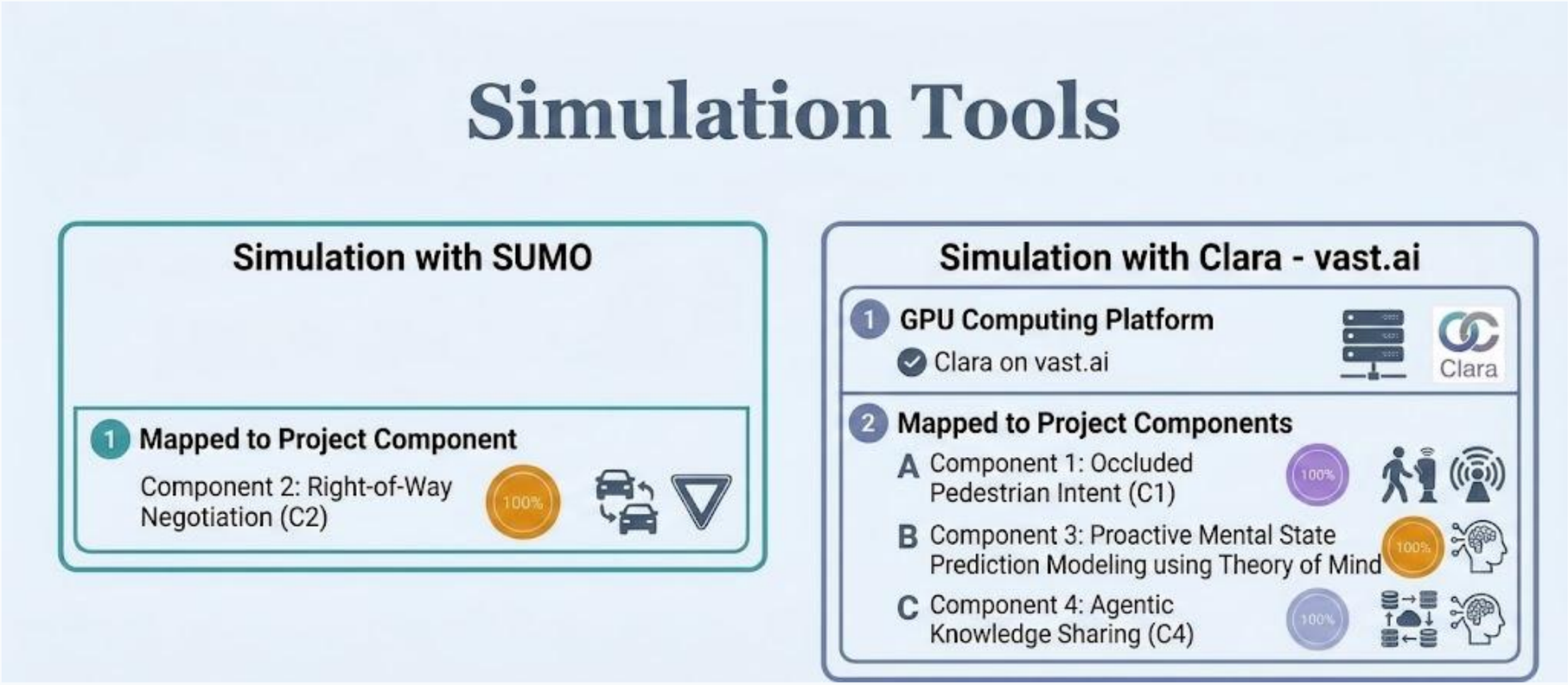
Limitations

- Limited number of vehicles and experimental scenarios
- Some classes showed temporary performance degradation during successive learning rounds
- Limited traffic-sign classes and environmental variations
- Limited computational resources constrained the scale of experiments

5. Integration, completion status



5. Integration, completion status



Component-wise NFR Handling

Because the four components are not a sequential dependency pipeline, NFR evidence is shown component-by-component and then defended under one AV-safety research theme.

Component	Safety + reliability	Explainability / responsibility	Privacy, deployment + evidence
C1 **Occlusion-aware pedestrian intent**	Defers unreliable cases with OBSERVE_MORE; track continuation only after prior visibility; calibrated uncertainty supports reliability.	Integrated Gradients explains evidence groups: intent, uncertainty, reliability and motion.	PIE test evidence: 1,152 sequences; 87.59% intent accuracy; 88.94% committed accuracy; honest detector/occlusion limits.
C2 **Multi-agent right-of-way negotiation**	Rules and conflict reasoning are applied first; MAPPO is used only for unresolved/cyclic cases; deterministic planner outputs READY/BLOCKED.	Directed precedence graph makes yield relations visible: A → B means A yields to B.	SUMO demo proves execution path; use final collision, deadlock, throughput and travel-time metrics here.
C3 **ToM-inspired pedestrian decision agent**	Most-critical pedestrian selected by minimum TTC; decisions are MAINTAIN, SLOW_DOWN or STOP using TTC risk.	Neuro-symbolic design: neural mental-state inference + rule-based decision with plain-language reason.	Leakage checks, 30-frame windows, LSTM/GRU ensemble and class-imbalance handling support model validity.
C4 **Selective knowledge sharing for road signs**	Global verification accepts/rejects shared embeddings before broadcast; local retraining updates only final CNN layer.	Agentic pipeline shows detection → share/ignore decision → verification → broadcast → retraining	Privacy by design: raw images are not shared — instead, embeddings are shared.

6. Commercialization, PLSSE, limitations and closing message

Who benefits?
AV/ADAS research teams, simulation labs, road-safety researchers, smart mobility education, and companies testing interpretable autonomous-driving modules.

Why adopt?
Safety-focused decision support, explainable outputs, privacy-preserving knowledge transfer, and reusable simulation/demo infrastructure.

How to sustain?
Pilot as a research/demo framework, extend datasets and simulation scenarios, and use evidence packs for validation before real-world deployment.

Professional / legal / social / security / ethical boundaries

1	**No overclaiming**	C1 cannot detect a fully invisible, never-observed pedestrian.
2	**Human oversight**	All outputs support AV decisions; they are not final real-road certification.
3	**Privacy and data use**	C4 shares compact embeddings + labels, not raw images.
4	**AI disclosure**	AI/model/code assistance must be verified and disclosed.

Final takeaways

1 Perception: handle occlusion and partial evidence
2 Reasoning: infer intentions, mental states and conflicts
3 Decision: negotiate or defer when certainty is low
4 Learning: share verified compact knowledge safely

Closing line for the panel: “Our project proves four independent but complementary AV intelligence capabilities: understanding uncertainty, explaining decisions, negotiating safely, and sharing useful knowledge without exposing raw data.”